Smart Automatic Plant Watering System Using ESP32

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Abstract

The ESP32 Soil Moisture Monitoring and Automatic Plant Watering System is a smart irrigation system designed to monitor the moisture level of soil and provide water to plants automatically when required. The system uses a **soil moisture sensor** to continuously detect the moisture content of the soil. The sensor data is processed by an **ESP32 microcontroller**, which controls the water pump based on the moisture level.

When the soil becomes dry, the ESP32 activates the pump and supplies water to the plant. Once the required moisture level is reached, the pump is automatically switched off. This helps prevent both overwatering and underwatering, while reducing water wastage and the need for manual watering.

INTRODUCTION

Water is an essential requirement for healthy plant growth, and proper irrigation is important for maintaining adequate soil moisture. Conventional plant watering methods generally require regular manual monitoring and watering, which may lead to overwatering or underwatering.

The **Smart Automatic Plant Watering System Using ESP32** is designed to automate the watering process based on the moisture level of the soil. A **soil moisture sensor** continuously detects the moisture content and sends the corresponding reading to the **ESP32 Development Board**. The ESP32 processes the sensor data and compares it with a predefined moisture threshold.

When the soil becomes dry, the ESP32 activates a **relay module**, which turns ON the **water pump** to supply water to the plant. Once the soil reaches the required moisture level, the ESP32 turns OFF the pump automatically. The system also uses the **Serial Monitor** to display real-time soil moisture readings and system status.

This automatic approach helps reduce manual effort, avoid unnecessary water usage, and maintain suitable moisture conditions for plants.

Apparatus Required

Hardware Components

- Esp32 Development Board
- Soil Moisture Sensor
- Water Pump
- Relay Module
- Water Container
- Connecting Wires / Jumper Wires
- Breadboard
- USB Cable
- Water Pipe / Tube
- Power Supply

Software Components

- Arduino IDE
- ESP32 Board Package
- Embedded C/C++
- Serial Monitor

Esp32 Development Board



Purpose : Main controller that processes sensor data and controls the pump.

Dual-core 32-bit microcontroller with built-in Wi-Fi and Bluetooth; 12-bit ADC for reading analog sensor values; 3.3V logic; multiple GPIO pins for connecting sensors and output devices; programmed through USB using the Arduino IDE.

Soil Moisture Sensor



Purpose : Measures the moisture level in the soil.

Sensor used to detect the water content of the soil; provides an analog output corresponding to the moisture condition; connected to an analog input of the ESP32; the sensor reading is compared with a predefined threshold to determine whether the soil is dry or wet.

Water Pump



Purpose : Pumps water to the plant when the soil is dry.

Small DC water pump used to transfer water from the water container to the plant; operates when the ESP32 detects a dry-soil condition; controlled indirectly through the relay module; helps provide automatic irrigation without manual watering.

Relay Module



Purpose : Switches the water pump ON/OFF using the ESP32.

Electrically controlled switching device used between the ESP32 and water pump; receives a control signal from the ESP32; switches the pump ON when the soil is dry and OFF when sufficient moisture is detected; provides isolation between the controller and the pump circuit.

Water Container



Purpose : Stores water for irrigation.

Container used as the water reservoir for the automatic watering system; provides the water required by the pump; the pump draws water from the container and transfers it through the water tube to the plant.

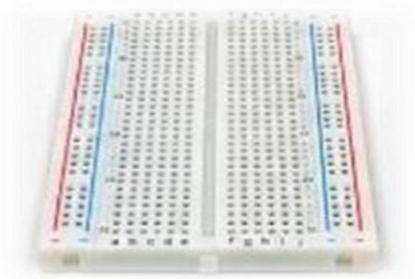
Connecting Wires / Jumper Wires



Purpose : Used for electrical connections.

Flexible electrical wires used to connect the ESP32 with the soil moisture sensor, relay module, LEDs, buzzer and other circuit components; available in male-to-male, male-to-female and female-to-female types depending on the connection required.

Breadboard



Purpose : Used to build and test the circuit.

Solderless prototyping board used for assembling the electronic circuit; provides interconnected holes for inserting components and jumper wires; allows the circuit to be modified and tested easily without permanent soldering.

USB Cable



Purpose : Provides power and uploads the program to ESP32.

USB cable used to connect the ESP32 development board to a computer; provides power to the ESP32 during development and allows the control program to be uploaded from the Arduino IDE to the board.

Water Pipe / Tube



Purpose : Carries water from the pump to the plant.

Flexible tube connected to the outlet of the water pump; transfers the pumped water from the water container to the plant; provides a controlled path for delivering water directly to the soil.

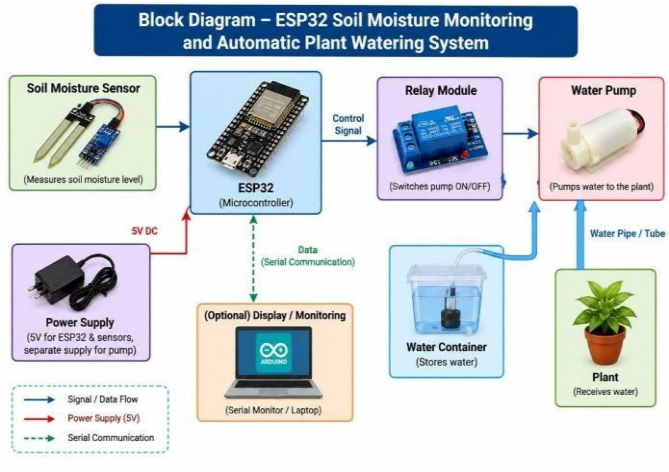
Power Supply



Purpose : Provides the required power to the circuit and pump.

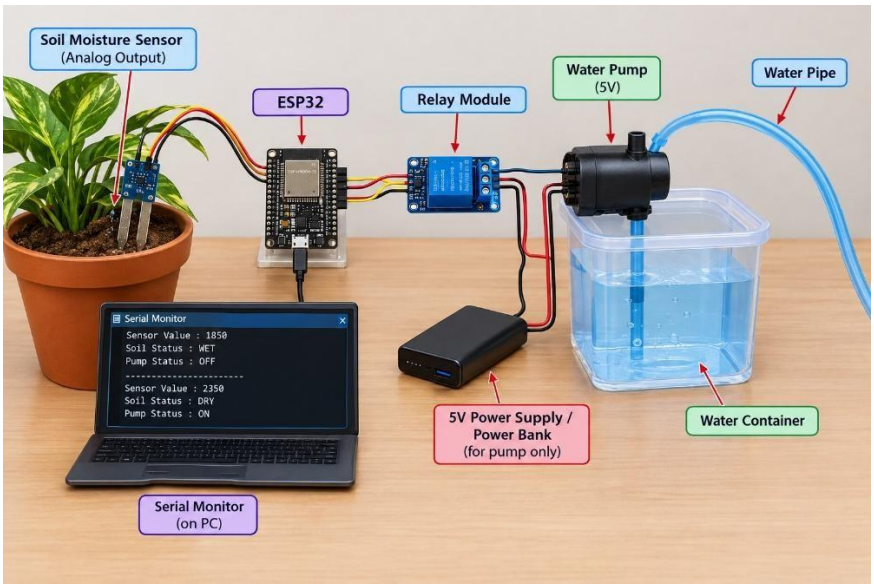
External power source used to provide sufficient electrical power for the system; supplies the required power for the water pump and supporting circuit components; an external supply is useful because the ESP32 board alone should not be used to power a pump.

Block Diagram



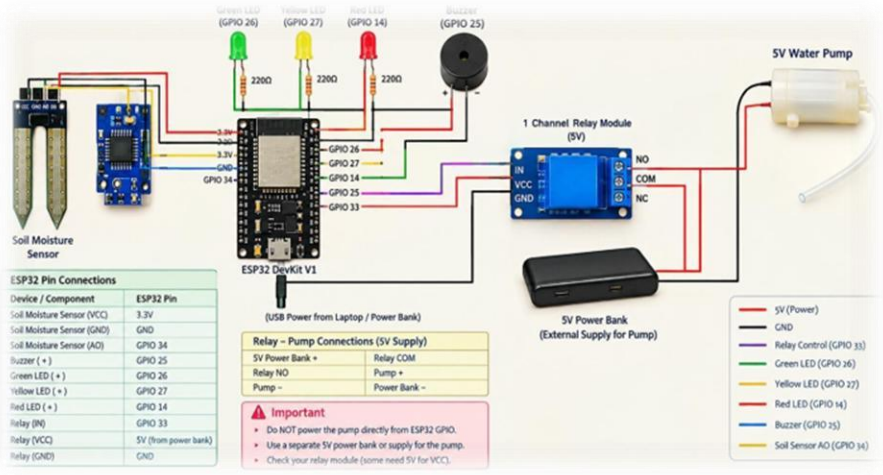
- **Soil Moisture Sensor:** Detects the moisture level present in the soil and provides the corresponding sensor reading to the ESP32.
- **ESP32 Development Board:** Processes the soil moisture data and determines whether the soil is dry or sufficiently moist.
- **Relay Module:** Acts as an electronic switch to control the **ON/OFF operation of the water pump** based on the ESP32 output.
- **Water Pump:** Automatically supplies water to the plant when the soil moisture level falls below the required threshold.
- **Water Tank:** Stores the water required for automatic irrigation and supplies it to the pump.
- **Power Supply:** Provides the necessary electrical power to operate the ESP32, sensors, relay, and water pump.
- **Serial Monitor:** Displays real-time soil moisture readings and the current system status, such as **DRY, MOIST, PUMP ON,** or **PUMP OFF**.
- **Working Principle:** The system continuously monitors the soil moisture level. When the soil becomes **dry**, the ESP32 activates the relay and turns **ON the water pump**. Water is supplied to the plant until the soil reaches the required moisture level. Once the soil becomes **moist**, the ESP32 deactivates the relay and turns **OFF the water pump** automatically.

Working Principle



- The **soil moisture sensor** is inserted into the soil of the plant and gives an analog voltage that changes with the amount of water present in the soil.
- The **ESP32** reads this analog value on GPIO 34 using its ADC and compares it with a fixed **threshold value of 2000**.
- If the reading is **greater than the threshold**, the soil is **DRY**. The ESP32 sends a control signal to the **relay module**, the relay closes its contact and the **water pump is switched ON**.
- The pump draws water from the **water container** and delivers it through the **water pipe** to the plant.
- When the reading falls to or below the threshold, the soil is **WET**. The relay is turned OFF and the **pump stops automatically**.
- The sensor value, soil status and pump status are printed on the **Serial Monitor** every **2 seconds**.
- The pump is powered from a **separate 5V supply / power bank**, so the ESP32 pins are never loaded by the pump current

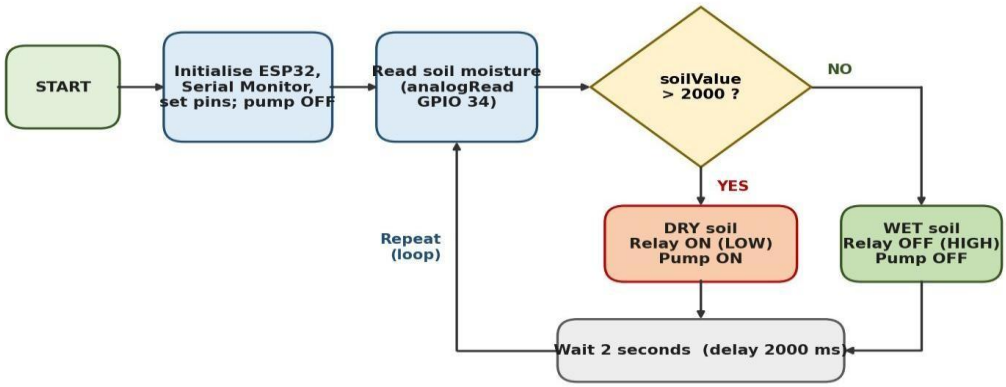
Circuit Diagram



- The circuit consists of an **ESP32 development board** connected to a **soil moisture sensor, three LEDs, a buzzer, a relay module, and a 5V water pump**.
- The **soil moisture sensor** is inserted into the soil near the plant roots. It detects the amount of moisture in the soil and produces an **analog output voltage** corresponding to the moisture level.
- The analog output of the soil moisture sensor is connected to **GPIO 34 of the ESP32**. GPIO 34 is an **ADC (Analog-to-Digital Converter) input**, allowing the ESP32 to read the sensor's analog signal as a numerical value.
- The ESP32 continuously reads the sensor value and compares it with a predefined **threshold value of 2000**.
- **Sensor value > 2000 → Soil is DRY**
- **Sensor value ≤ 2000 → Soil is WET**
- The **three LEDs** provide visual indication of the system condition:
- **Green LED** – indicates that the soil has sufficient moisture and the pump is OFF.
- **Yellow LED** – indicates the system/monitoring condition.
- **Red LED** – indicates that the soil is dry and watering is required.

Soil Moisture Sensor → ESP32 ADC (GPIO 34) → Compare with Threshold 2000 → ESP32 → Relay → 5V Pump → Water supplied to Plant

Flowchart



- **Start**
  Begins the automatic plant watering process. The ESP32 starts the monitoring system.
- **Initialize ESP32**
  Sets pins and starts the Serial Monitor. The water pump is initially kept OFF.
- **Read Soil Moisture**
  Reads the soil moisture value from GPIO 34. The reading is checked continuously.
- **Check Soil Value > 2000?**
  The ESP32 compares the moisture value with 2000. This determines whether the soil is dry or wet.
- **Dry Soil**
  If the value is above 2000, the relay turns ON. The water pump starts watering the plant.
- **Wet Soil**
  If the value is below 2000, the relay turns OFF. The water pump stops watering.
- **Wait 2 Seconds**
  The system waits for 2000 milliseconds.
  Then it repeats the moisture-reading process.

Repeat Loop
Soil moisture is monitored continuously.
The pump operates automatically when required

Advantages

1. Saves Water

The system supplies water only when the soil moisture level falls below the required level, helping to reduce unnecessary water usage.

2. Prevents Overwatering and Underwatering

The automatic control of the pump helps maintain the required moisture level and reduces the risk of excessive or insufficient watering.

3. Reduces Manual Effort

The system automatically monitors soil moisture and controls the water pump, reducing the need for frequent manual watering.

4. Low Cost and Easy to Build

The system uses simple and readily available components, making it economical and easy to assemble.

5. Real-Time Monitoring

The **Serial Monitor** displays soil moisture readings and system status, allowing the user to observe the operation in real time.

Limitations

1. Sensor Corrosion

The sensor probe may corrode over time when continuously exposed to wet soil.

2. Fixed Moisture Threshold

The fixed threshold value may not be suitable for every type of plant or soil condition.

3. Separate Power Supply for Pump

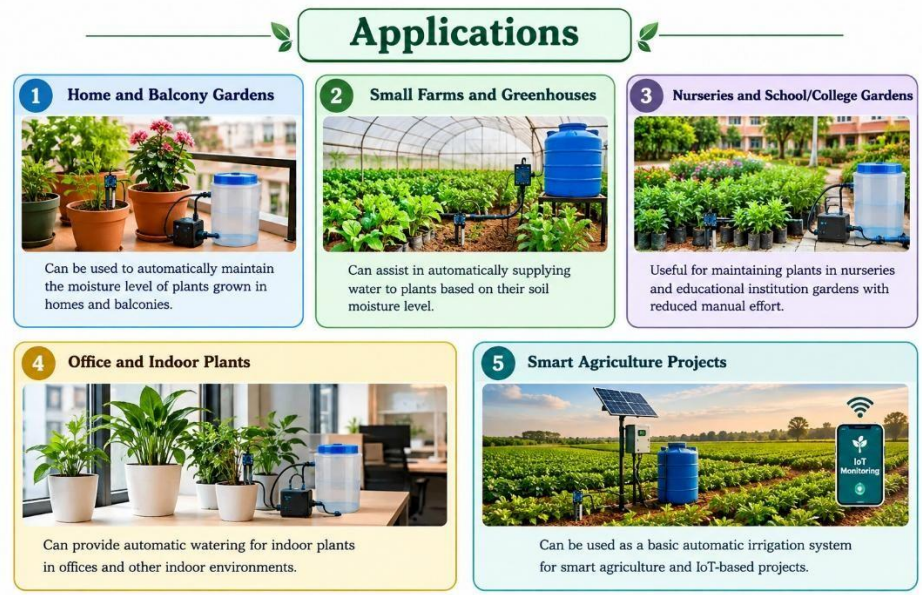
The water pump requires a suitable power supply in addition to the controller's power source.

4. Manual Water Tank Refilling

The water container must be manually refilled when the available water is exhausted.

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Applications



1. Home and Balcony Gardens

Can be used to automatically maintain the moisture level of plants grown in homes and balconies.

2. Small Farms and Greenhouses

Can assist in automatically supplying water to plants based on their soil moisture level.

3. Nurseries and School/College Gardens

Useful for maintaining plants in nurseries and educational institution gardens with reduced manual effort.

4. Office and Indoor Plants

Can provide automatic watering for indoor plants in offices and other indoor environments.

5. Smart Agriculture Projects

Can be used as a basic automatic irrigation system for smart agriculture and IoT-based projects.

Future Scope



1. Wi-Fi/IoT-Based Monitoring

The ESP32 can be connected to Wi-Fi to enable remote monitoring and control through a mobile application.

2. Corrosion-Resistant Capacitive Sensor

A capacitive soil moisture sensor can be used to improve durability and reduce corrosion-related issues.

3. Water-Level Monitoring

A water-level sensor can be added to the tank to detect low water levels and provide an alert to the user.

4. Solar-Powered Operation

Solar power can be integrated to provide an environmentally friendly and independent power source for the system.

5. Weather-Based and Scheduled Watering

Future versions can incorporate weather conditions and predefined schedules to improve the efficiency of automatic irrigation.

code

```
// ESP32 Automatic Plant Watering System
// Pin connections follow the ESP32 connection table

#define SOIL_SENSOR   34   // Soil moisture sensor AO
#define GREEN_LED     18   // via 220 ohm resistor
#define YELLOW_LED    19   // via 220 ohm resistor
#define RED_LED       21   // via 220 ohm resistor
#define BUZZER        23   // Buzzer signal (+)
#define RELAY_PIN     22   // Relay module IN

// Most 1-channel relay modules are active-LOW (LOW = relay ON).
// If your pump behaves opposite, change this to false.
#define RELAY_ACTIVE_LOW true

int threshold = 2000;      // > 2000 = DRY, <= 2000 = WET (calibrate for your sensor)

void pumpOn() {
  digitalWrite(RELAY_PIN, RELAY_ACTIVE_LOW ? LOW : HIGH);
}

void pumpOff() {
  digitalWrite(RELAY_PIN, RELAY_ACTIVE_LOW ? HIGH : LOW);
}

void beep(int times, int duration) {
  for (int i = 0; i < times; i++) {
    digitalWrite(BUZZER, HIGH);
    delay(duration);
    digitalWrite(BUZZER, LOW);
    delay(duration);
  }
}
```

```
void setup() {
  Serial.begin(115200);

  pinMode(SOIL_SENSOR, INPUT);
  pinMode(GREEN_LED, OUTPUT);
  pinMode(YELLOW_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);
  pinMode(RELAY_PIN, OUTPUT);

  // Everything OFF at start (pump OFF first)
  pumpOff();
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(RED_LED, LOW);
  digitalWrite(BUZZER, LOW);

  Serial.println("ESP32 Automatic Plant Watering System");
  Serial.println("--------------------------------------");
}

void loop() {
  // Yellow LED blinks briefly on every reading (monitoring indicator)
  digitalWrite(YELLOW_LED, HIGH);

  int soilValue = analogRead(SOIL_SENSOR);

  Serial.print("Soil Moisture Value: ");
  Serial.println(soilValue);

  if (soilValue > threshold) {
    // DRY soil -> pump ON
    Serial.println("Soil Status : DRY");
    Serial.println("Pump Status : ON");
```

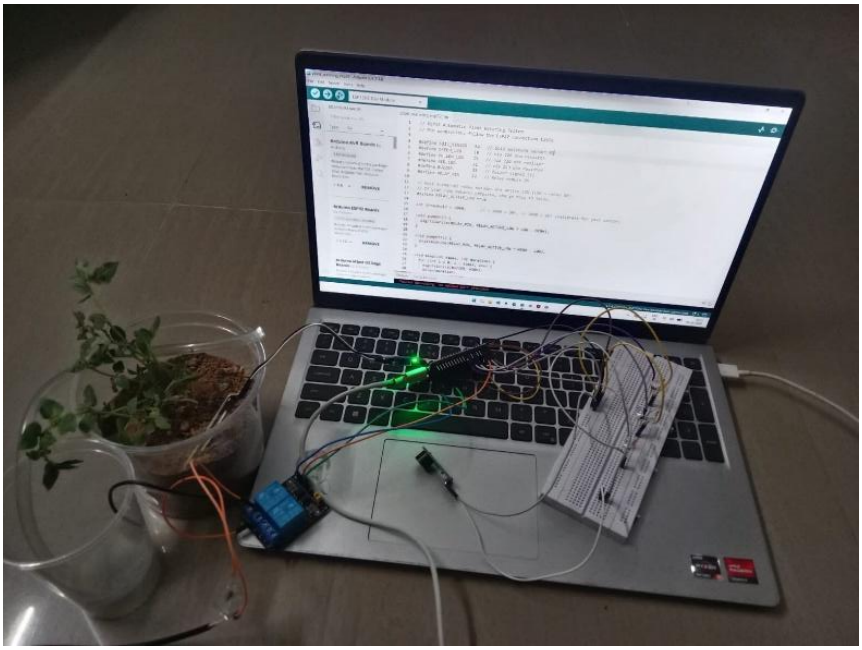
```
    digitalWrite(RED_LED, HIGH);
    digitalWrite(GREEN_LED, LOW);
    pumpOn();
    beep(2, 100);          // short double beep as alert
  } else {
    // WET soil -> pump OFF
    Serial.println("Soil Status : WET");
    Serial.println("Pump Status : OFF");

    digitalWrite(GREEN_LED, HIGH);
    digitalWrite(RED_LED, LOW);
    pumpOff();
    digitalWrite(BUZZER, LOW);
  }

  Serial.println("-------------------------------------");

  delay(200);
  digitalWrite(YELLOW_LED, LOW);
  delay(1600);             // total loop time is about 2 seconds
}
```

Output



ESP32 Automatic Plant Watering System

Soil Moisture Value: 2500

Soil Status : DRY

Pump Status : ON

Result and Observation

Condition	Soil Status	Relay	Pump	Serial Monitor
Sensor value > 2000 (e.g. 2500)	DRY	ON (LOW)	ON	Soil Status : DRY / Pump Status : ON
Sensor value ≤ 2000	WET	OFF (HIGH)	OFF	Soil Status : WET / Pump Status : OFF

- The sample output shows a reading of **2500**, which is above the threshold of 2000, so the system reports **DRY** soil and switches the **pump ON**.
- As water reaches the soil, the reading falls, the status changes to **WET** and the pump switches **OFF** automatically.
- The Serial Monitor prints the moisture value and the pump status every 2 seconds, so the system can be monitored easily.
- The system works without manual intervention and avoids both **overwatering** and **underwatering**.

Conclusion

The Smart Automatic Plant Watering System using ESP32 successfully monitors the soil moisture level and controls the water pump automatically. When the soil is dry the pump is switched ON, and when the required moisture is reached it is switched OFF. The system reduces water wastage, avoids overwatering and underwatering, and removes the need for manual watering. It is simple, low cost and can be extended with IoT features in the future.